

YANGHANYU ZHAO

+1 (680) 697-3795 | zhaoyanghanyu@gmail.com | [LinkedIn](#)

EDUCATION

Syracuse University

Master of Science in Computer Science

Syracuse, NY

2023 – 2025

Dalian Neusoft University of Information

Bachelor of Engineering in Software Engineering

Dalian, Liaoning

2018 – 2022

WORK EXPERIENCE

Neusoft Group (Dalian) Co., Ltd

Dalian, China

Java Software Engineer

July 2021 – August 2023

- Launched more than 10 production releases (Dec 2022, Aug 2023) of insurance policy and calculation modules for Sumitomo Life Insurance (one of Japan's largest life insurers with 14M+ policyholders), engineering backend **REST APIs** with **Java Spring Boot** and **AWS** (S3, Lambda, Kinesis) for scalable document storage, serverless processing, and real-time clickstream analysis; designed and optimized **MySQL** database schemas for policy and claims data, and used **Redis** for caching to improve query performance.
- Elevated customer satisfaction from 4.2 to 4.6 out of 5 by leading a systematic quality improvement initiative across 90 design books (60 external, 30 internal): analyzed client-reported defects to identify high-frequency bug patterns, developed targeted unit and functional test suites using **JUnit** and **Mockito** to prevent regressions, and authored 60 standardized testing books to enable consistent testing practices across the team.
- Optimized internal tooling across the full stack — revamped data computation modules in **Java** and redesigned 10 documentation pages using **TypeScript**, **React**, and **CSS**, improving usability for internal teams.

Yunnan Yitu Technology Co., Ltd

Kunming, China

Software Engineer (Intern)

June 2020 – December 2020

- Developed a full-stack power monitoring system: built a real-time visualization interface in **React** and **Redux** to display power flow diagrams, fault alarms, and performance metrics, and designed **RESTful APIs** to enable data exchange between the control software and external applications.

AI EXPERIENCE

Multi-Model Toxic Comment Classification and Evaluation

- Built an end-to-end **NLP pipeline** to classify **toxic social media comments**, including text cleaning, tokenization, stop-word removal, stemming/lemmatization, **EDA**, **feature engineering**, model training, and evaluation.
- Engineered **TF-IDF**, **TextBlob sentiment polarity**, and **Doc2Vec** feature representations to capture lexical importance, emotional tone, and semantic context in user-generated comments.
- Trained and compared **Naive Bayes**, **Logistic Regression**, **SVM**, and **Keras Sequential Neural Network** models; evaluated performance with **accuracy**, **precision**, **recall**, **F1-score**, confusion matrices, **ROC curves**, and **AUC**.
- Identified **Logistic Regression with TF-IDF/sentiment features** as the strongest model, achieving about **88% accuracy** and **0.94 AUC**; analyzed **feature-combination trade-offs** and observed that Doc2Vec features did not consistently improve model performance.

Machine Learning Model Implementation and Evaluation

- Implemented **linear/logistic regression**, **neural networks**, **random forests**, **PCA**, and **K-Means** using **Python**, **PyTorch**, **scikit-learn**, NumPy, pandas, and Matplotlib.
- Performed **preprocessing**, **feature scaling**, **train/test splitting**, **validation analysis**, and **model comparison** across classification, regression, clustering, and dimensionality-reduction tasks.

LLM-Enhanced Microservice Commerce Platform

March 2026 – April 2026

- Built an **LLM-enabled software system** that connected natural-language interaction with backend workflows for product search, order lookup, and checkout.
- Integrated the assistant into a **distributed microservice architecture** using JWT-secured APIs, OpenFeign, and Kafka for synchronous and event-driven coordination.
- Emphasized **reliability and control** by separating LLM-based intent interpretation from deterministic business execution, and packaged the system with Docker Compose for reproducible end-to-end deployment.

SOFTWARE ENGINEERING PROJECT

Gmail | Distributed Systems · Microservice Architecture · E-Commerce Platform

July 2025 – Present

- Architected a distributed e-commerce platform with 10+ microservices using **Spring Boot** and **Spring Cloud (Nacos)** for service discovery and configuration, **Feign** for inter-service calls, **Sentinel** for circuit breaking, **Sleuth + Zipkin** for distributed tracing), with **API Gateway** handling routing, rate limiting, and authentication.
- Implemented **Elasticsearch** for full-text product search with multi-field filtering, sorting, keyword highlighting, and paginated results, replacing traditional MySQL LIKE queries to enable sub-second search across the full product catalog. Integrated **RabbitMQ** for asynchronous order processing and inventory deduction, leveraging dead-letter queues as delay queues to auto-cancel unpaid orders after timeout, and implemented publisher confirm + manual ACK mechanisms to guarantee reliable message delivery and ensure eventual consistency across services.
- Built responsive storefronts using **Vue.js** and **ElementUI** for product catalog, cart, checkout, and user authentication; managed data persistence with **MySQL** and **Redis** caching to reduce API response latency.
- Containerized all 10+ microservices with **Docker**, orchestrated deployment on **Kubernetes** with auto-scaling and rolling updates, and built a **Jenkins CI/CD** pipeline integrating Git-triggered builds, automated unit testing, Docker image publishing, and zero-downtime deployment to production.

ContentGuard – Blockchain-based NFT Marketplace

February 2025 – June 2025

- Built a decentralized NFT marketplace on **Ethereum** (Sepolia testnet), enabling users to mint, buy, and sell NFTs with verified copyright and on-chain ownership via Solidity smart contracts (ERC-721); developed a responsive **React.js** frontend integrated with **Web3.js** and **MetaMask** for wallet authentication, live transaction feedback, and a dynamic auction dashboard; leveraged **IPFS** for immutable content and metadata storage.

SKILLS

Languages: Java, Python, JavaScript, TypeScript, SQL, HTML/CSS.

Software Engineering: Spring Boot, Spring Cloud, RESTful APIs, Microservices, React, Vue.js, Web3.js.

AI/ML/NLP: scikit-learn, TensorFlow/Keras, NLTK, TextBlob, Gensim, TF-IDF, Doc2Vec, Logistic Regression, SVM, Naive Bayes.

QA & Evaluation: JUnit, black-box testing, white-box testing, unit testing, model evaluation, error analysis, confusion matrix, ROC/AUC, F1-score.

Tools & Infrastructure: MySQL, Redis, Elasticsearch, RabbitMQ, AWS, Docker, Kubernetes, Jenkins, Git, Linux.